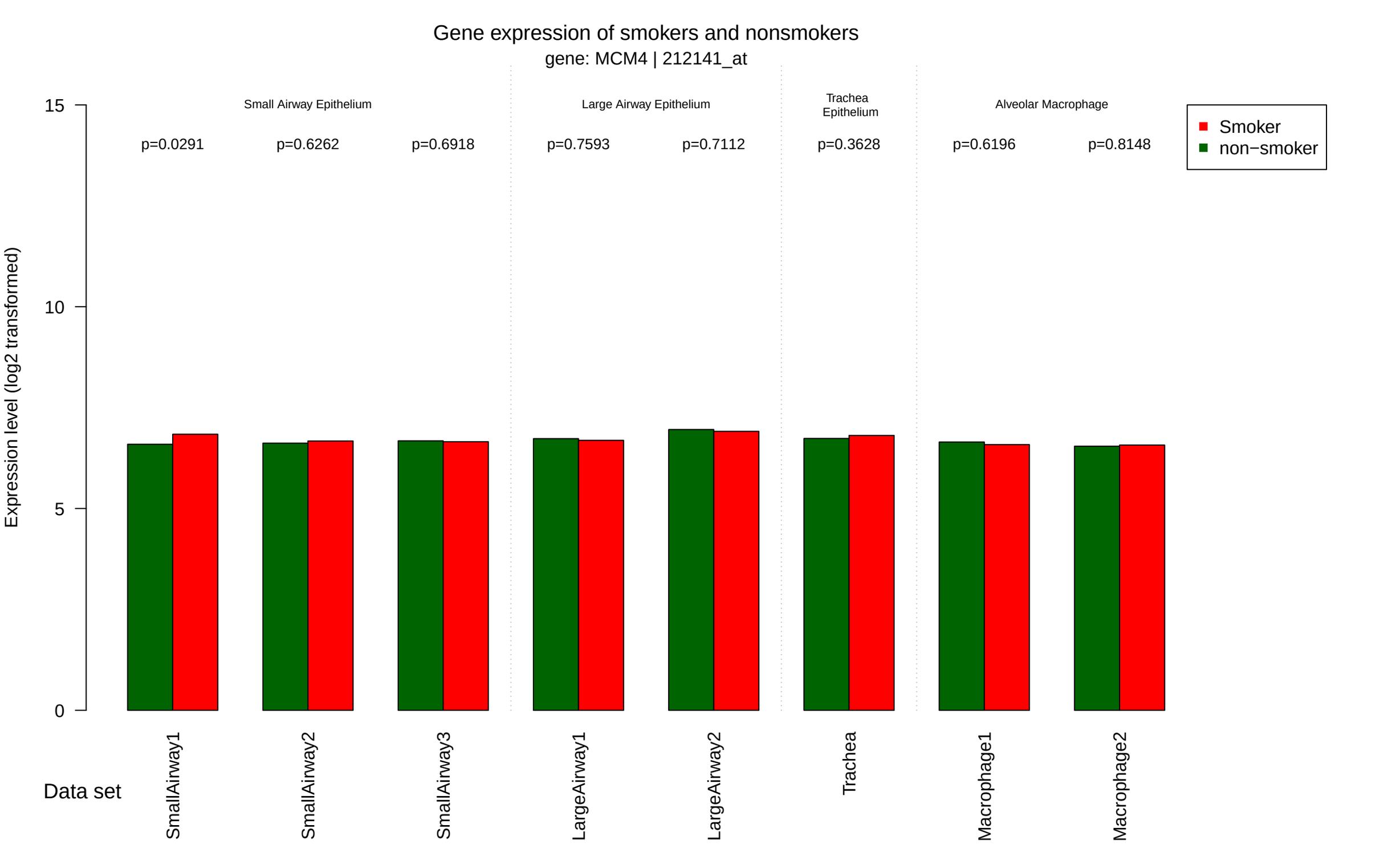
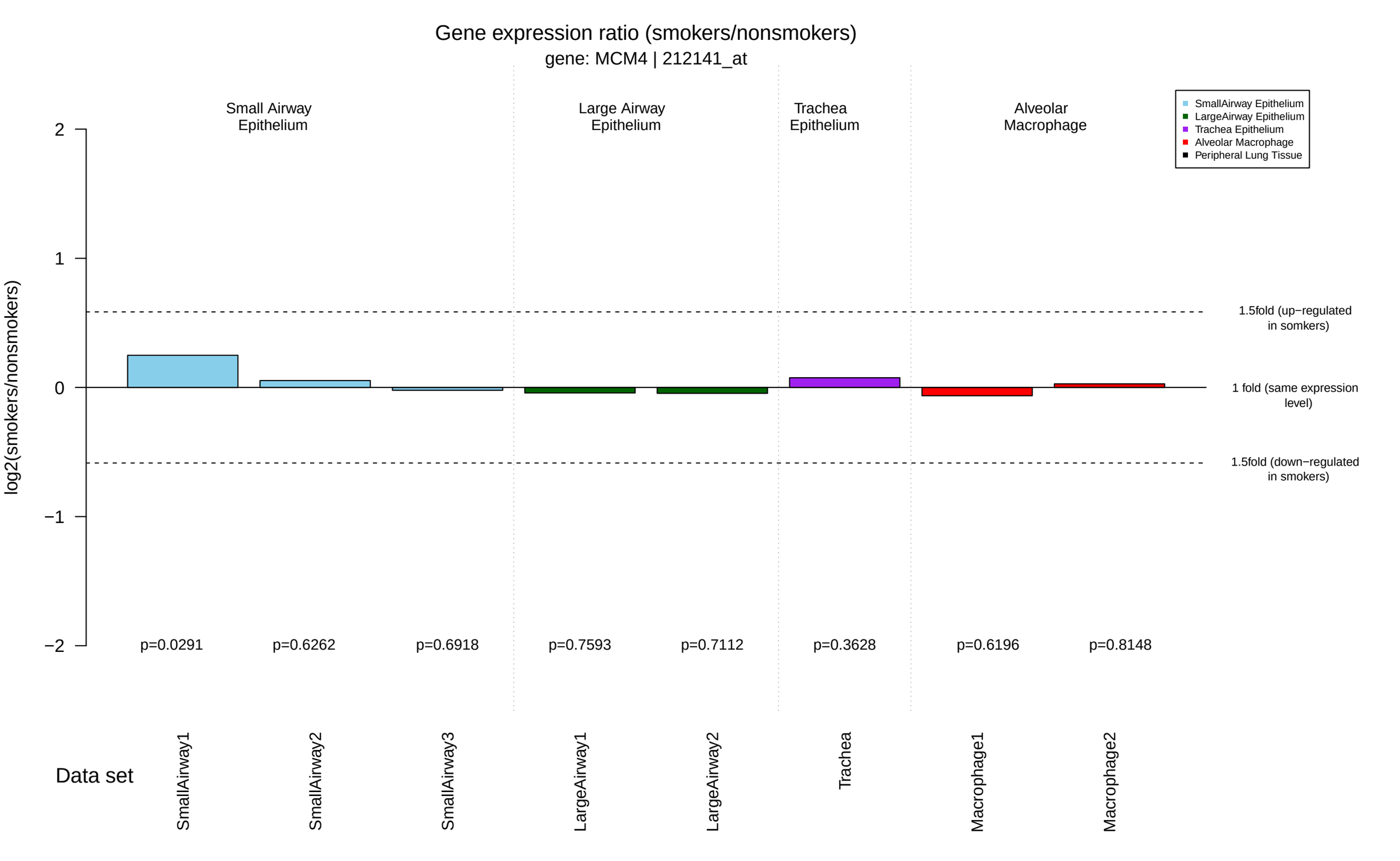
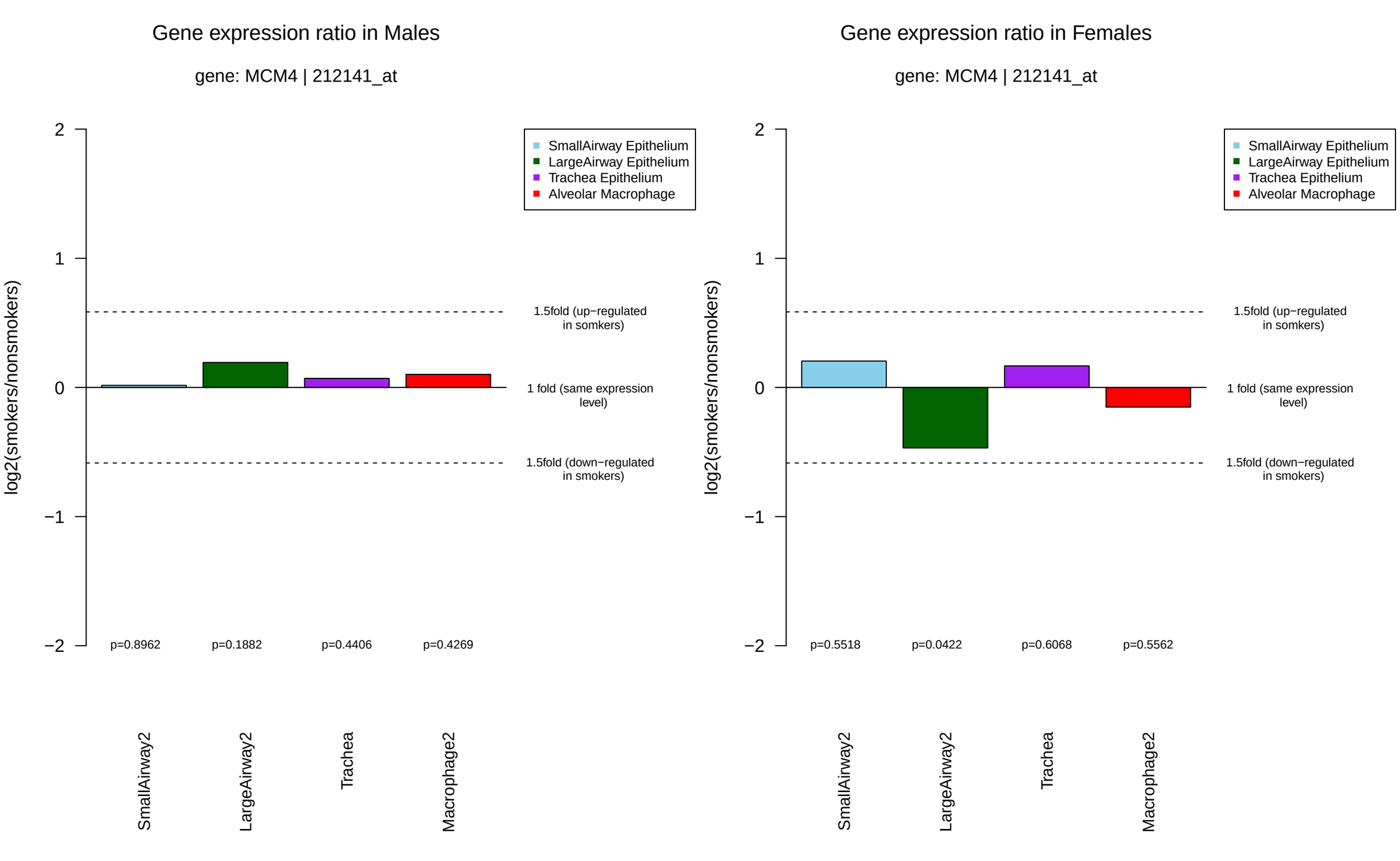
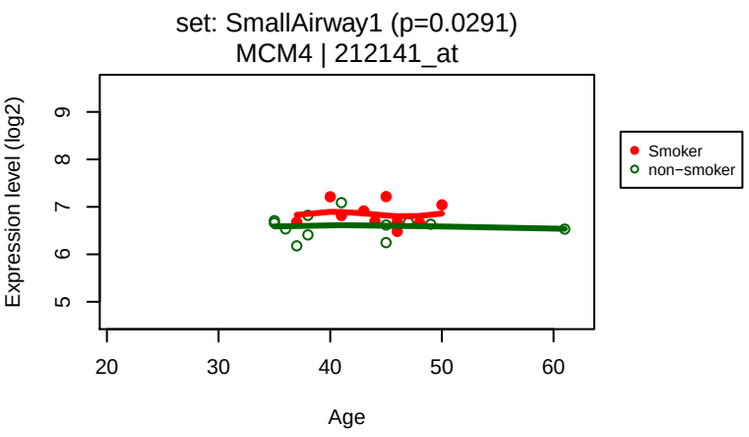
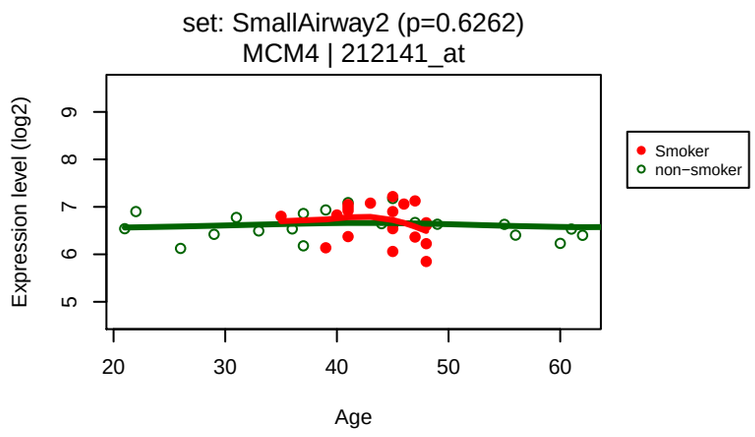
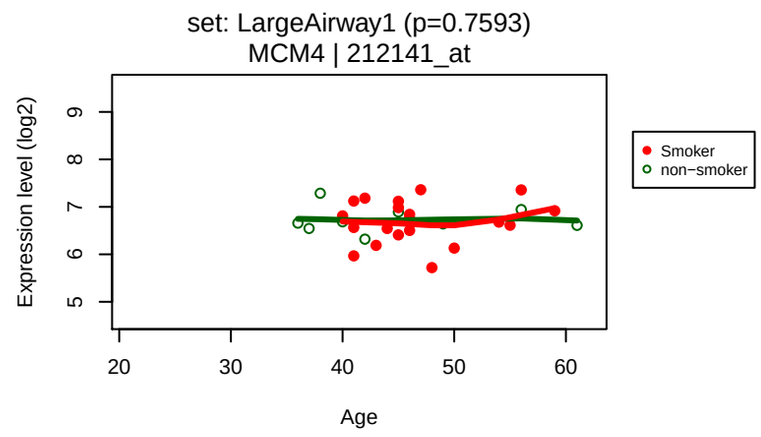

Gene expression of smokers and nonsmokers
gene: MCM4 | 212141_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0291
p=0.6262
p=0.6918
p=0.7593
p=0.7112
p=0.3628
p=0.6196
p=0.8148
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

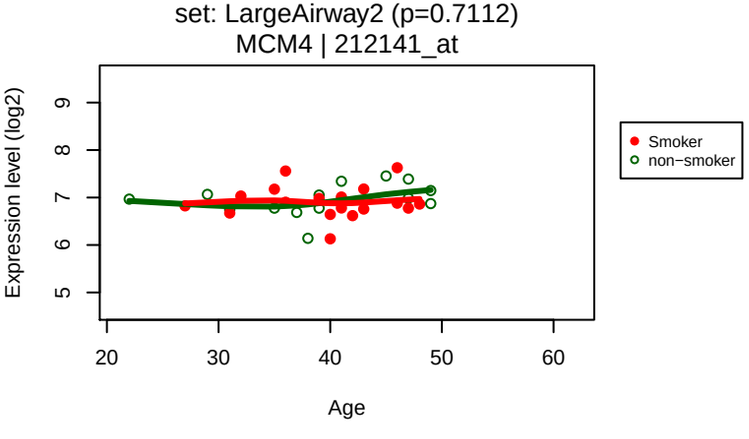

Gene expression ratio (smokers/nonsmokers)
gene: MCM4 | 212141_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0291
p=0.6262
p=0.6918
p=0.7593
p=0.7112
p=0.3628
p=0.6196
p=0.8148
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

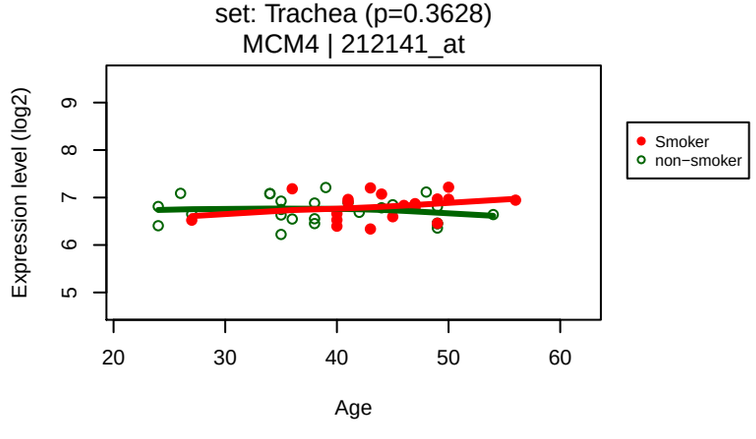

Gene expression ratio in Males
gene: MCM4 | 212141_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8962
p=0.1882
p=0.4406
p=0.4269
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MCM4 | 212141_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5518
p=0.0422
p=0.6068
p=0.5562
SmallAirway2
LargeAirway2
Trachea
Macrophage2

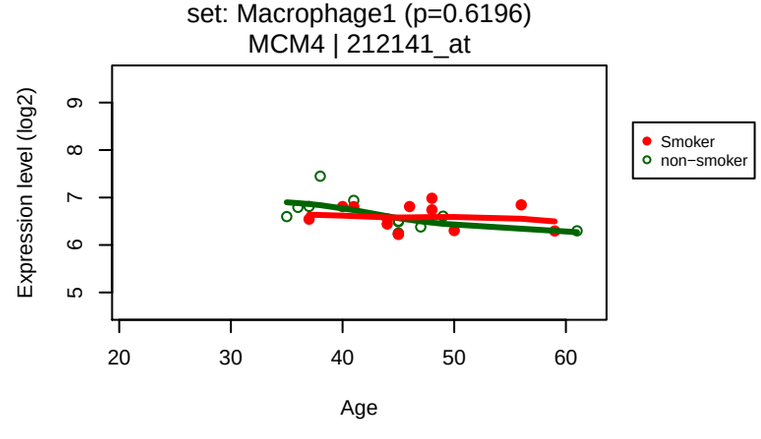

set: SmallAirway1 (p=0.0291)
MCM4 | 212141_at
Expression level (log2)
Age
Smoker
non−smoker

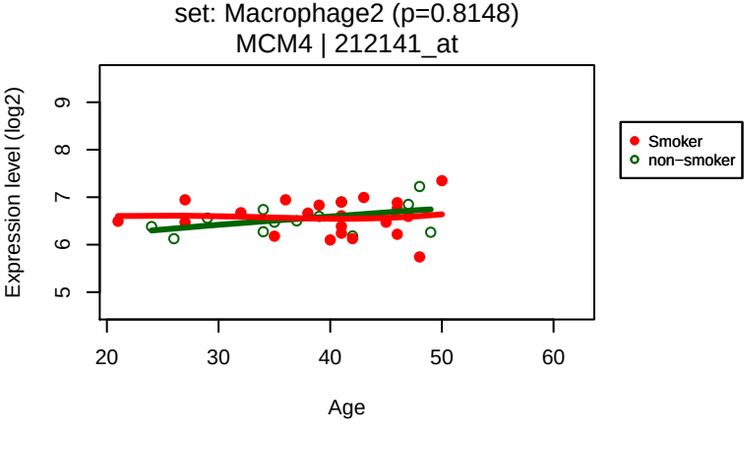

set: SmallAirway2 (p=0.6262)
MCM4 | 212141_at
Expression level (log2)
Age
Smoker
non−smoker

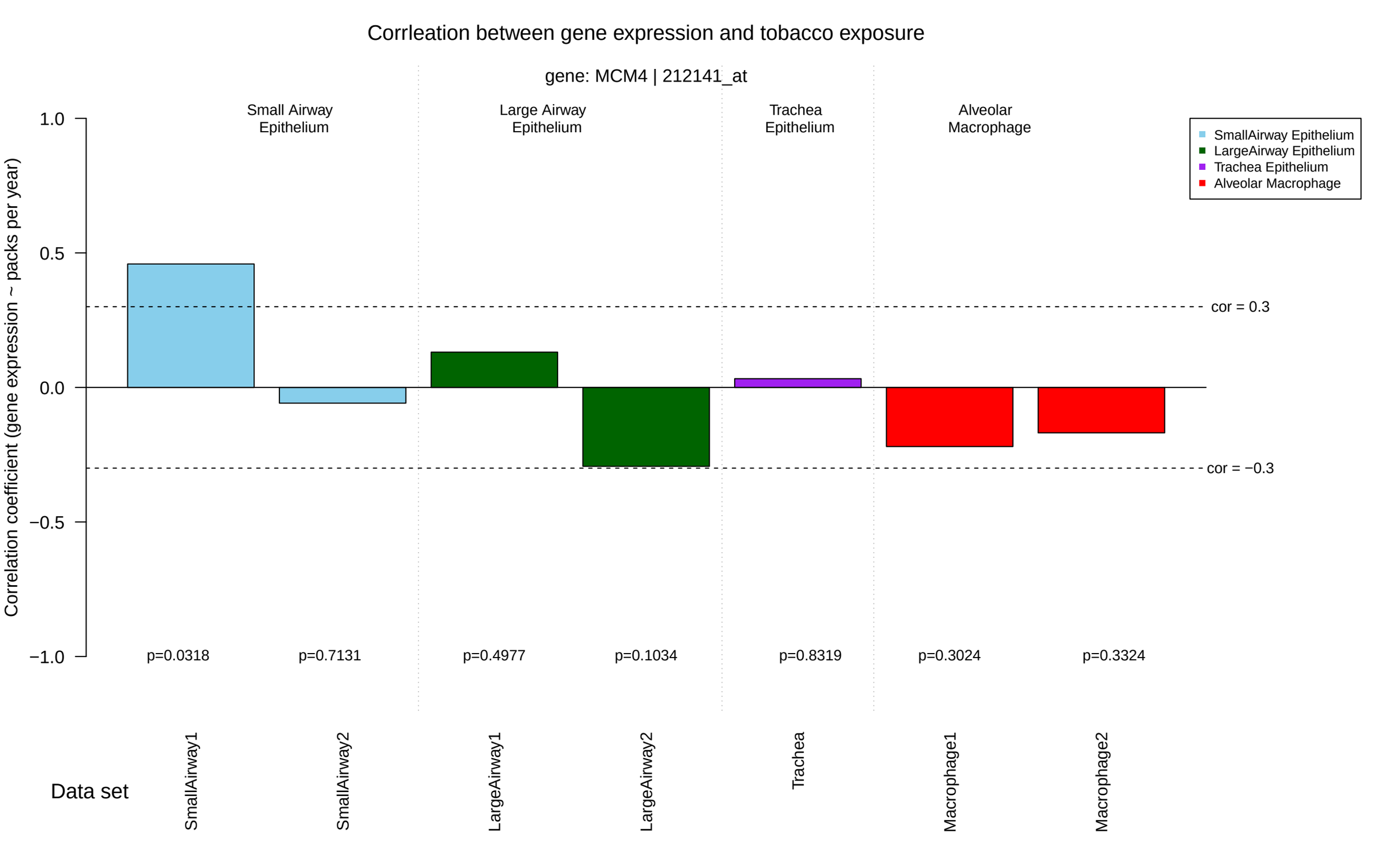

set: LargeAirway1 (p=0.7593)
MCM4 | 212141_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7112)
MCM4 | 212141_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3628)
MCM4 | 212141_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6196)
MCM4 | 212141_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8148)
MCM4 | 212141_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MCM4 | 212141_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0318
p=0.7131
p=0.4977
p=0.1034
p=0.8319
p=0.3024
p=0.3324
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2